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Paper

All-Dimensional Coherence in Higher λ-Models: Front-Seed Semantics, Proof-Relevant Witnesses, and the K_∞ Model

Daniel O. Martínez-Rivillas,[1] Arthur F. Ramos[2,*] and Ruy J.G.B. de Queiroz[1]

[1]Centro de Informática (CIn), Universidade Federal de Pernambuco (UFPE), Recife, Pernambuco, Brazil
[2]Microsoft, USA

[*]Corresponding author. arfreita@microsoft.com

Abstract

Martínez-Rivillas and de Queiroz established a semantic framework in which untyped λ-terms are interpreted in extensional Kan complexes, higher $\beta\eta$-conversions are organized into an explicit tower of n-cells, and the inclusion $\mathrm{TH}_{\lambda=} \subseteq \mathrm{HoTFT}$ holds for every such complex. In a companion paper they constructed the K_∞ homotopy λ-model as an inverse limit of iterated function spaces and exhibited a nontrivial separation phenomenon at the level of individual witnesses. Taking those two developments as a closed baseline, the present paper fills four specific gaps left by the existing framework. First, *all-dimensional constructive coherence*: the higher λ-conversion tower carries an all-dimensional constructive globular coherence package extending a recursively completed low-dimensional λ-ω-groupoid core, with strict source/target compatibility at every level. Second, *reduced front-seed semantic coherence*: a bare extensional Kan complex together with exactly two explicit 3-cell data—a whiskering-left/whiskering-right comparison and an inner-right-front pentagon contraction—suffices to recover the recursive associator comparison theorem and the semantic pentagon identity. Third, *proof-relevant K_∞ witness semantics*: on the fixed witness-separation span, every explicit 1-term witness receives a canonical K_∞ interpretation determined by its β/η class, and the resulting semantic separation persists uniformly through every dimension of the higher-cell tower, including the recursively completed dimensions. Fourth, *exact K_∞ reflexive packaging*: the inverse limit K_∞ carries globally continuous reify, reflect, and application maps satisfying explicit stagewise formulas, yielding a complete reflexive c.h.p.o. The exact reflexive packaging theorem is the most concrete model-specific result; the other theorems clarify how much of the higher structure is forced once the low-dimensional witness language, the reduced coherence data, and the point-level β/η separation are fixed. Taken together, these results separate the genuinely model-specific input from the structural consequences of the recursive higher tower. All results have been formally verified in Lean 4.

Keywords: higher lambda-calculus, proof relevance, extensional Kan complexes, omega-groupoid, denotational semantics, formal verification

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1. Introduction

1.1. Background and lineage

The untyped λ-calculus has served since its creation as a meeting ground for logic, computation, and algebra. The classical denotational tradition, beginning with Scott's construction of continuous lattices [12] and codified in Barendregt's [1] systematic treatment, interprets λ-terms as elements of reflexive domains—ordered structures carrying reify and reflect maps. In that tradition the key equation $F(G(f))(x) = f(x)$ (and its dual) collapses the distinction between data and function, and the resulting models are *extensional* precisely when $G(F(x)) = x$. This interplay between the logical structure of convertibility and the computational content of reduction witnesses is a central theme of the present paper.

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A complementary thread, motivated by the homotopy-theoretic programme of Grothendieck [4] and the algebraic coherence machinery of Batanin [2], asks whether the conversion relation of a logical calculus can itself be organized into a higher-categorical object—a weak ω-groupoid whose n-cells are explicit n-dimensional conversion witnesses. In the setting of dependent type theory, that question was answered affirmatively by the development of homotopy type theory [5], where identity types naturally form a weak ω-groupoid.

The work of Martínez-Rivillas and de Queiroz connects these two traditions for the *untyped* calculus. In a series of papers [8, 11, 10, 9] they introduced a framework in which λ-terms are interpreted in *extensional Kan complexes*, higher $\beta\eta$-conversions are organized into a tower of explicit n-cells, and the classical conversion theory is recovered as the 0-truncation of a richer homotopical structure.

A further ingredient in this circle of ideas is the computational-paths programme of de Veras, Ramos, de Queiroz, and de Oliveira [3], which demonstrated that the paths generated by directed reduction steps in a rewriting system carry natural weak groupoid structure. That work, published in this journal [3], established a direct link between directed rewriting and higher-categorical algebra. That observation provides a syntactic precursor to the semantic picture developed here: the explicit reduction sequences between λ-terms play the role of computational paths, and the higher-dimensional conversion witnesses are the coherence data for the resulting higher groupoid.

1.2. The two baseline papers

We rely on two specific antecedents as a closed baseline; all results quoted from them are taken as established.

Paper I [10] introduced extensional Kan semantics for the untyped λ-calculus, defined the explicit tower of higher β- and η-conversions through dimension 3, and proved the central witness-lifting theorem: for every extensional Kan complex K, every explicit reduction witness lifts to a semantic 1-cell, and explicit syntactic 2- and 3-cells lift to semantic 2- and 3-cells between the corresponding interpreted witnesses. The proof proceeds by induction on the structure of reduction sequences, lifting individual-step β- and η-soundness to a full semantic interpretation of each dimension of the explicit tower. The resulting theory enriches the classical proposition-level equality $M =_{\beta\eta} N$ with a proof-relevant layer of semantic n-cells. Paper I also established $\beta\eta$-confluence via Hindley–Rosen and proved the invertibility of reduction sequences via Church–Rosser.

Paper II [9] constructed the K_∞ homotopy λ-model as the inverse limit $K_\infty = \varprojlim_n K_n$ of a tower of iterated function spaces over a nontrivial homotopy-group base $K_0 = \mathbb{N}^+$. The construction follows the classical Scott tradition but enriches it with an explicit homotopical layer. Paper II proved that K_∞ is an algebraic bounded-complete domain (Proposition 4.1) and exhibited a separation phenomenon, recalled below in Example 8.1: the β-contraction and the η-contraction of a common redex land at the same target but receive semantically distinct images in K_∞. That example established the nontriviality of the proof-relevant structure but did not analyze the witness language at the level of explicit reduction terms or trace the separation through the recursively completed higher-cell tower.

1.3. Contributions of the present paper

Starting from the closed baseline described above, this paper fills four specific gaps left by Papers I and II. The four theorem packages are not all of the same kind. Section 7.2 contains the most concrete model-specific result, Section 6 isolates the exact coherence interface needed later, and Sections 5 and 8 are structural results describing how the recursive tower behaves once the low-dimensional or point-level data are fixed.

Main contributions.

(A) **All-dimensional constructive coherence (Section 5).** We prove a structural completion theorem for the explicit higher λ-conversion tower. Paper I supplies explicit data through dimension 3, and the recursive higher-derivation machinery already suggests how to continue above that range. The content of Theorem 5.6 is to compare the explicit low-dimensional presentation with the equality-generated continuation, identify the intermediate dimensions in which those presentations meet, and prove strict globular boundary compatibility for the resulting all-dimensional tower.

(B) **Reduced front-seed semantic coherence (Section 6).** We isolate the precise refinement of the coherent semantic interface needed later: a bare extensional Kan complex equipped with a whiskering-left/whiskering-right comparison and only the inner-right-front pentagon contraction already suffices to recover the recursive associator comparison theorem, the semantic pentagon identity for interpreted reduction sequences, and the source/target/shell bridge theorems. The gain is that the full pentagon witness can be replaced by this smaller front-facing seed while keeping WLWR explicit.

(C) **Proof-relevant K_∞ witness separation (Section 8).** We refine the point-level separation of Paper II on the witness-separation span of Example 8.1 to a theorem about explicit witnesses on that fixed span. The main new input is a canonical classification of explicit 1-witnesses into β- and η-classes together with their canonical K_∞ interpretations. Once that point-level dichotomy is in place, the higher-dimensional non-connection statements become structural consequences of the equality-generated tower.

(D) **Exact K_∞ reflexive packaging (Section 7.2).** We establish that the inverse limit K_∞ carries globally continuous reify, reflect, and application operations satisfying explicit stagewise formulas, yielding a complete reflexive c.h.p.o. with a strictly stronger retract law ($h(k(x)) = x$ rather than merely $h(k(x)) \leq x$) than the standard domain-theoretic formulation requires.

1.4. Structure of the paper

Sections 2–4 collect definitions and restate the baseline results of Papers I and II. Sections 5–8 contain the new theorems, presented with full proofs. Section 9 gives a detailed comparison with related work and a list of open problems.

Sections 5 and 8 should be read as structural completion and persistence theorems: they explain what follows once the low-dimensional core and the point-level K_∞ separation are fixed. Sections 6 and 7.2 isolate the semantic interface and the model-specific inverse-limit structure on which those structural arguments rest.

The mathematical content comes first throughout. The formal development [6] plays an evidential role: a brief verification note at the end of the paper records that all stated results have been mechanically checked, but the paper is self-contained and can be read without reference to the proof assistant. The constructive character of the proofs—particularly the explicit witness constructions in Theorems 5.6 and 8.5—reflects the paper's orientation toward the logic-and-computation tradition, where the computational content of proofs is a first-class object of study.

2. Preliminaries and Setup

2.1. Lambda-terms and reduction

We work with the untyped λ-calculus using de Bruijn indices. A *term* is generated by:

$$M, N ::= \texttt{var}\ n \mid M\ N \mid \lambda\, M$$

where $n \in \mathbb{N}$ is a de Bruijn index. We write $M\ N$ for application and $\lambda\, M$ for abstraction. Substitution $M[N]$ denotes the capture-free replacement of the variable at index 0 in M by N, with the standard shifting conventions.

A *β-step* is a single reduction of the form $(\lambda\, M)\, N \to_\beta M[N]$, closed under subterm contexts. An *η-step* is a single reduction of the de Bruijn form

$$\lambda\, ((\mathsf{shift}(1,0)M)\, (\texttt{var}\ 0)) \to_\eta M,$$

equivalently

$$\lambda\, (M\, (\texttt{var}\ 0)) \to_\eta \mathsf{shift}(-1,0)M,$$

when $\texttt{var}\ 0$ does not occur free in M. A *$\beta\eta$-step* is either a β-step or an η-step.

Definition 2.1 (Reduction sequences) A *reduction sequence* from M to N is a finite sequence of $\beta\eta$-steps and their formal inverses:

$$M = M_0 \leftrightarrow M_1 \leftrightarrow \cdots \leftrightarrow M_k = N.$$

We write $\mathrm{RedSeq}(M, N)$ for the type of all such sequences. Each reduction sequence is an explicit proof-relevant witness for the $\beta\eta$-convertibility of M and N.

2.2. Extensional Kan complexes

The semantic setting is that of *Kan complexes* in the simplicial sense.

Definition 2.2 (Kan complex) A *Kan complex* K is a simplicial set equipped with a chosen horn filler: for every horn $\Lambda \subset \Delta^{n+1}$ with a missing face, a chosen $(n+1)$-simplex σ extending the given faces. Formally, K carries a function

$$\mathsf{fill} : \mathrm{Horn}(K, n, j) \to K_{n+1}$$

satisfying the face-matching specification for all non-missing faces.

Definition 2.3 (Reflexive and extensional Kan complexes) A Kan complex K is *reflexive* if it is equipped with maps $F : |K| \to [K \to K]$ and $G : [K \to K] \to |K|$ satisfying the η-law $F(G(f))(x) = f(x)$. It is *extensional* if additionally $G(F(x)) = x$ for all objects x. Here $[K \to K]$ denotes the endofunction space on 0-simplices used by the semantic interpretation, and $|K|$ is the set of 0-simplices.

An extensional Kan complex K interprets λ-terms via:

$$[\![\texttt{var}\ n]\!]_\rho = \rho(n), \qquad [\![M\ N]\!]_\rho = F([\![M]\!]_\rho)([\![N]\!]_\rho), \qquad [\![\lambda\, M]\!]_\rho = G\big(x \mapsto [\![M]\!]_{\rho[0 \mapsto x]}\big).$$

Definition 2.4 (The theories $\mathrm{TH}_{\lambda=}$, $\mathrm{TheoryEq}_K$, and HoTFT) Fix an extensional Kan complex K. The *model-specific equality theory* is

$$\mathrm{TheoryEq}_K(M,N) \;:\equiv\; \forall\,\rho,\ [\![M]\!]_\rho = [\![N]\!]_\rho.$$

Separately, the *classical λ-theory* $\mathrm{TH}_{\lambda=}$ is the syntactic relation $M =_{\beta\eta} N$. The *homotopical theory* $\mathrm{HoTFT}(K)$ enriches $\mathrm{TheoryEq}_K$ to a proof-relevant tower: $\mathrm{HoTFT}_1(K)$ assigns semantic 1-cells between denotations, while $\mathrm{HoTFT}_2(K)$ and $\mathrm{HoTFT}_3(K)$ are indexed by parallel lower-dimensional witnesses.

Remark 2.5 (Comparing the three theories) The three notions should be kept distinct. The classical theory $\mathrm{TH}_{\lambda=}$ is syntactic and propositional: the assertion $M =_{\beta\eta} N$ records only that some finite conversion chain exists, without remembering any particular chain. For a fixed model K, the relation $\mathrm{TheoryEq}_K$ forgets witnesses in a different way: it asks only that all environments give equal denotations. By contrast, $\mathrm{HoTFT}(K)$ retains the *explicit witness*: a pair (M,N) in $\mathrm{HoTFT}_1(K)$ is accompanied by a specified semantic path, and the higher layers are indexed by parallel lower-dimensional witnesses. The passage from $\mathrm{TH}_{\lambda=}$ to HoTFT is therefore not merely a set-theoretic strengthening but a genuinely proof-relevant enrichment. This enrichment is not an artefact of the formalism. Theorem 8.5 below demonstrates that the K_∞ model separates β-witnesses from η-witnesses even when they connect the same pair of terms, so the additional structure carried by HoTFT is genuinely non-degenerate.

2.3. Higher conversions

Definition 2.6 (Explicit higher cells) Given terms M, N, a *2-cell* $\alpha : p \Rightarrow q$ between reduction sequences $p, q : \mathrm{RedSeq}(M,N)$ is a formal witness of homotopy between p and q, built inductively from reflexivity, symmetry, transitivity, congruence under whiskering, and the structural interchange, associator, pentagon, and triangle operations.

A *3-cell* $\Theta : \alpha \Rrightarrow \beta$ between 2-cells is built by the same schema one dimension higher.

The tower of n-cells is organized into *n-conversions* Σ_n (reduction sequences at dimension 1, homotopies at dimension 2, etc.) and *n-terms* Π_n (explicit witnesses for n-conversions).

3. Baseline: Soundness and $\mathrm{TH}_{\lambda=} \subseteq \mathrm{HoTFT}$

The results of this section are not new: they were established in Paper I [10] and are restated here in the notation of the present paper so that the subsequent sections can cite them precisely. The reader already familiar with [10] may safely skip to Section 4.

The foundation of the semantic theory is the soundness of β- and η-reduction, and the architecture by which that foundation lifts through the explicit tower is worth stating precisely, both because it fixes notation and because the subsequent sections extend the same inductive strategy to higher-dimensional data.

The logical structure of the lifting is as follows. At the base, one proves that a single β-step (respectively η-step) induces a semantic 1-cell in any extensional Kan complex; this is the content of the two soundness theorems below. A key supporting ingredient is the *substitution lemma*: the semantic interpretation commutes with syntactic substitution,

$$[\![M[N/0]]\!]_\rho \;=\; [\![M]\!]_{\rho[0\mapsto[\![N]\!]_\rho]},$$

which reduces soundness of the β-rule to a computation in the function space. From these individual-step results one passes to composite reduction sequences by induction on sequence length: concatenation of sequences lifts to composition of semantic paths, and formal inversion lifts to symmetry of paths.

At the next level, each inductive constructor of a 2-cell—left whiskering, right whiskering, horizontal composition, and the associator, unitor, and interchange 2-cells—is shown to lift to a semantic 2-cell, using the composition and whiskering operations of the Kan complex. The 3-cell constructors (pentagon, triangle, and the higher interchange witnesses) then lift by the same scheme one dimension higher, using the 3-dimensional horn-filling structure of the Kan complex to produce the required higher homotopies. In each case the inductive step is structural: the semantic image of a composite constructor is the corresponding composite of the semantic images of its constituents.

The three theorems below record the cumulative outcome of this lifting procedure.

Theorem 3.1 (β-soundness) For every extensional Kan complex K, every β-step $M \to_\beta N$ induces a semantic path from $[\![M]\!]_\rho$ to $[\![N]\!]_\rho$ for every environment ρ.

Theorem 3.2 (η-soundness) For every extensional Kan complex K, every η-step $M \to_\eta N$ induces a semantic path from $[\![M]\!]_\rho$ to $[\![N]\!]_\rho$ for every environment ρ.

By induction on the structure of reduction sequences and their higher-cell constructors, these individual-step soundness results lift to witness-level interpretations of the explicit tower:

Theorem 3.3 ($\mathrm{TH}_{\lambda=} \subseteq \mathrm{HoTFT}$, $\Sigma_2 \subseteq \mathrm{HoTFT}_2$, $\Sigma_3 \subseteq \mathrm{HoTFT}_3$) Let K be an extensional Kan complex. Then:

(i) Every reduction sequence $p : \mathrm{RedSeq}(M, N)$ induces a semantic path in K. In particular, every syntactic $\beta\eta$-conversion witness in $\mathrm{TH}_{\lambda=}$ admits a proof-relevant semantic lift in $\mathrm{HoTFT}_1(K)$.
(ii) If $p, q : \mathrm{RedSeq}(M, N)$ and $\alpha : p \Rightarrow q$, then α induces a semantic 2-cell between the interpreted 1-cells of p and q.
(iii) If $p, q : \mathrm{RedSeq}(M, N)$, $\alpha, \beta : p \Rightarrow q$, and $\Theta : \alpha \Rrightarrow \beta$, then Θ induces a semantic 3-cell between the interpreted 2-cells of α and β.

These results are exactly those established in [10]. We take them as the starting point for the new theorems presented below.

4. Higher Conversions and the Explicit Tower

4.1. The globular tower of n-conversions

The n-conversion tower is a *globular set*: at dimension 0 one has terms, at dimension 1 reduction sequences between terms, at dimension 2 homotopies between reduction sequences, at dimension 3 higher homotopies between homotopies, and so on. Source and target maps $s, t : \Sigma_{n+1} \to \Sigma_n$ record the boundary of each cell, and the globularity conditions $s \circ s = s \circ t$ and $t \circ s = t \circ t$ hold by construction.

Through dimensions 0–3, the cells are built by explicit inductive types carrying the full combinatorial structure of λ-reduction: whiskering, horizontal composition, associators, pentagon and triangle coherences, interchange, and so on. At dimension 4 and above, the cells are generated by a uniform *recursive completion* principle.

4.2. The omega-groupoid core

The low-dimensional structure through dimension 3 forms a weak ω-groupoid core. Concretely, the following data are available:

- *Composition*: concatenation of reduction sequences.
- *Inverses*: formal reversal of reduction sequences.
- *Whiskering*: left and right whiskering of 2-cells along 1-cells.
- *Horizontal composition*: horizontal composition of 2-cells.
- *Associator*: a 2-cell $\alpha_{p,q,r} : (p \cdot q) \cdot r \Rightarrow p \cdot (q \cdot r)$.
- *Unitors*: 2-cells $\ell_p : \mathrm{id} \cdot p \Rightarrow p$ and $\rho_p : p \cdot \mathrm{id} \Rightarrow p$.
- *Pentagon*: a 3-cell witnessing the standard pentagon identity for four composable 1-cells.
- *Triangle*: a 3-cell witnessing the triangle identity relating the associator and the unitors.
- *Interchange*: 3-cells witnessing both parenthesizations of horizontal composition.

All whiskering operations preserve identity, composition, symmetry, and congruence up to explicit 3-cells.

Definition 4.1 (ω-groupoid on higher λ-conversions) The *λ-ω-groupoid* is the structure $\mathscr{G}_\lambda$ with objects = Term, 1-cells = RedSeq, 2-cells = $\mathrm{Homotopy}_2$, 3-cells = $\mathrm{Homotopy}_3$, and the above operations, satisfying the globularity conditions at all low dimensions.

5. All-Dimensional Constructive Coherence

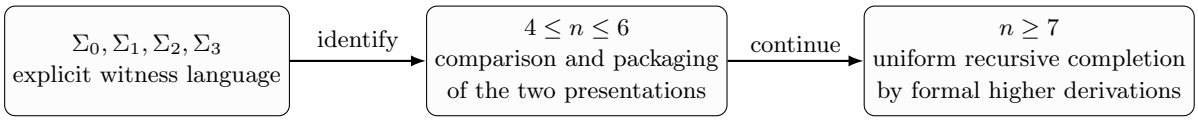
5.1. Recursive completion of the tower

The weak ω-groupoid $\mathscr{G}_\lambda$ constructed from explicit $\beta\eta$-paths already gives concrete data in low dimensions:

$$\Sigma_0 = \mathrm{Term}, \qquad \Sigma_1 = \mathrm{RedSeq}, \qquad \Sigma_2 = \mathrm{Homotopy}_2, \qquad \Sigma_3 = \mathrm{Homotopy}_3.$$

Moreover, the low-dimensional core also carries the next layers of reflexive higher coherence needed to iterate the construction. The point of this section is therefore structural rather than model-specific: once this finite core is fixed, no new primitive coherence operations are needed in higher dimension, and one can continue the tower uniformly by freely adding formal reflexivity, symmetry, and transitivity witnesses. The non-formal part of the argument is to show that this equality-generated continuation agrees with the explicit low-dimensional presentation and preserves strict globular boundaries. Figure 1 summarizes the three phases of that comparison.

Remark 5.1 (Which layers are explicit) Two different presentations are in play here. The syntactic tower $\Sigma_\bullet$ is explicit through dimension 3, and for $n \geq 4$ its cells are already given recursively as higher-derivation data between parallel $(n-1)$-cells. Separately, the shared simplicial core used by the generic coherence packaging is a reflexive 6-cell core: its 4-, 5-, and 6-cells are packaged reflexivity layers attached to the low-dimensional λ-ω-groupoid interface. The role



$\mathscr{G}_\lambda$ acquires an all-dimensional constructive globular coherence package
with strict source/target compatibility at every level.

Figure 1 Architecture of the proof of Theorem 5.6. The explicit low-dimensional tower is first matched with the equality-generated continuation in the bridge dimensions, after which the recursive completion proceeds uniformly.

of dimensions 4–6 in Theorem 5.6 is precisely to identify these two presentations before the uniform recursive continuation begins above dimension 6.

Definition 5.2 (Higher derivation) Let X be a type. The type $\mathrm{HigherDeriv}_X(x, y)$ of *higher derivations* between $x, y \in X$ is generated inductively by

$$\mathsf{refl}_x : \mathrm{HigherDeriv}_X(x, x), \qquad \mathsf{symm} : \mathrm{HigherDeriv}_X(x, y) \to \mathrm{HigherDeriv}_X(y, x),$$

and

$$\mathsf{trans} : \mathrm{HigherDeriv}_X(x, y) \to \mathrm{HigherDeriv}_X(y, z) \to \mathrm{HigherDeriv}_X(x, z).$$

Thus $\mathrm{HigherDeriv}_X$ is the reflexive–symmetric–transitive closure of the equality data already present on X.

The type $\mathrm{HigherDeriv}_X(x, y)$ does not introduce new primitive generators between unrelated objects. In particular, any element of $\mathrm{HigherDeriv}_X(x, y)$ determines that the endpoints agree. Accordingly, above the explicit core the recursive tower is *equality-generated*: higher cells are added by iterating formal reflexivity, symmetry, and transitivity on previously constructed parallel cells. Its central structural property is still *functoriality*: every map on the underlying type transports higher derivations covariantly.

Proposition 5.3 (Functoriality of HigherDeriv) Let $f : X \to Y$ be any map. There is a canonical map

$$\mathrm{HigherDeriv}(f) : \mathrm{HigherDeriv}_X(x, y) \longrightarrow \mathrm{HigherDeriv}_Y\big(f(x),\, f(y)\big),$$

defined by recursion on the constructors of HigherDeriv:

$$\begin{aligned}
\mathrm{HigherDeriv}(f)(\mathsf{refl}_x) &= \mathsf{refl}_{f(x)}, \\
\mathrm{HigherDeriv}(f)(\mathsf{symm}(h)) &= \mathsf{symm}\big(\mathrm{HigherDeriv}(f)(h)\big), \\
\mathrm{HigherDeriv}(f)(\mathsf{trans}(h_1, h_2)) &= \mathsf{trans}\big(\mathrm{HigherDeriv}(f)(h_1),\, \mathrm{HigherDeriv}(f)(h_2)\big).
\end{aligned}$$

Moreover, the assignment $f \mapsto \mathrm{HigherDeriv}(f)$ respects identity and composition:

$$\mathrm{HigherDeriv}(\mathrm{id}_X) = \mathrm{id}, \qquad \mathrm{HigherDeriv}(g \circ f) = \mathrm{HigherDeriv}(g) \circ \mathrm{HigherDeriv}(f).$$

Proof The recursive clauses define $\mathrm{HigherDeriv}(f)$ by structural induction on derivation terms. The identity law is immediate: $\mathrm{HigherDeriv}(\mathrm{id})$ fixes every constructor pointwise, since $\mathsf{refl}_x \mapsto \mathsf{refl}_{\mathrm{id}(x)} = \mathsf{refl}_x$, $\mathsf{symm}(h) \mapsto \mathsf{symm}(\mathrm{HigherDeriv}(\mathrm{id})(h))$, and the induction hypothesis closes the recursion.

For the composition law, the base case is:

$$\begin{aligned}
\mathrm{HigherDeriv}(g)(\mathrm{HigherDeriv}(f)(\mathsf{refl}_x)) &= \mathrm{HigherDeriv}(g)(\mathsf{refl}_{f(x)}) \\
&= \mathsf{refl}_{g(f(x))} \\
&= \mathrm{HigherDeriv}(g \circ f)(\mathsf{refl}_x).
\end{aligned}$$

The inductive cases for symm and trans follow by the same structural unwinding: the outer map $\mathrm{HigherDeriv}(g)$ distributes across the constructors produced by $\mathrm{HigherDeriv}(f)$, and the induction hypothesis matches each sub-derivation. □

Proposition 5.3 is the engine that drives the realization theorem below: at each dimension, the realization of a higher derivation is obtained by applying HigherDeriv to the endpoint realization map. The identity and composition laws ensure that this process is compatible with the globular boundary structure and with the recursive definition of source and target.

Starting from the reflexive 6-cell core of $\mathscr{G}_\lambda$, we define a recursive globular tower $\mathscr{R}_\bullet(\mathscr{G}_\lambda)$ as follows. For dimensions $0, \ldots, 6$ we keep the given low-dimensional cells of the core explicitly. For every $n \geq 6$, an $(n+1)$-cell is a triple

$$(x, y, h), \qquad x, y \in \mathscr{R}_n(\mathscr{G}_\lambda), \quad h \in \mathrm{HigherDeriv}(x, y).$$

Its source is x and its target is y. The globularity equalities are then automatic: if h is a higher derivation from x to y, the two iterated sources and the two iterated targets agree because every constructor of HigherDeriv preserves equality of endpoints.

This recursive completion should be compared with the explicit tower $\Sigma_\bullet$ of higher λ-conversions. The latter is already constructive in all dimensions, but its presentation is not literally the same as the generic recursive completion in dimensions $4, 5, 6$: there one must package the previously constructed 3-cells, 4-cells, and 5-cells as explicit cells of the target tower. Once these dimensions are handled, however, the higher stages on both sides are governed by the same rule: "a higher cell is a higher derivation between parallel cells of the preceding dimension."

The interface between the explicit low-dimensional tower and the recursive completion in the intermediate dimensions is captured by the following observation, which we state as a separate lemma because it isolates the only non-trivial step in passing from the low-dimensional core to the recursive continuation.

Lemma 5.4 (Low-dimensional packaging) For $d \in \{3, 4, 5\}$, let Σ_d denote the explicit d-cells of the λ-ω-groupoid, and let $\mathscr{R}_{d+1}(\mathscr{G}_\lambda)$ denote the $(d+1)$-cells of the recursive completion. There is a canonical *packaging map*

$$\mathsf{pack}_{d+1} : \mathscr{R}_{d+1}(\mathscr{G}_\lambda) \longrightarrow \Sigma_{d+1}$$

defined by:

(a) embedding the two endpoint d-cells into Σ_d via the identity (for $d \leq 3$) or via the inductively defined packaging map pack_d (for $d \in \{4, 5\}$), and then

(b) transporting the higher derivation datum along the resulting endpoint map, using the functoriality of HigherDeriv (Proposition 5.3).

No new primitive coherence operations beyond those already present in the 3-cell core enter the construction.

Proof At dimension 4, the endpoints are 3-cells, which belong to the explicit core and therefore need no packaging: pack_4 applies HigherDeriv(id) to the derivation datum, which by the identity law of Proposition 5.3 is the identity. Concretely, a 4-cell of the recursive completion consists of two parallel 3-cells η, θ together with a higher derivation $\omega : \mathrm{HigherDeriv}(\eta, \theta)$, and pack_4 sends this to the explicit 4-cell (η, θ, ω) verbatim.

At dimension 5, the endpoints are 4-cells. Each such 4-cell was itself produced by pack_4, so pack_5 applies $\mathrm{HigherDeriv}(\mathsf{pack}_4)$ to the derivation data. The composition law of Proposition 5.3 guarantees that iterated packaging respects the boundary maps. The argument for dimension 6 is identical, with pack_5 replacing pack_4.

The key observation is that every constructor invoked—refl, symm, trans, and the endpoint maps—already exists in the 3-cell core. The packaging maps merely *compose* these existing operations through the functoriality of HigherDeriv; they do not introduce any new generating data. □

5.2. The realization theorem

Definition 5.5 (All-dimensional coherence package) Let $\mathscr{C}$ be a reflexive low-dimensional globular core and let $\mathscr{T}$ be a globular tower. An *all-dimensional coherence package* from $\mathscr{C}$ to $\mathscr{T}$ consists of maps

$$\mathrm{realize}_n : \mathscr{R}_n(\mathscr{C}) \longrightarrow \mathscr{T}_n \qquad (n \in \mathbb{N}),$$

where $\mathscr{R}_\bullet(\mathscr{C})$ is the recursive completion of $\mathscr{C}$, such that for every $(n+1)$-cell x,

$$s(\mathrm{realize}_{n+1}(x)) = \mathrm{realize}_n(sx), \qquad t(\mathrm{realize}_{n+1}(x)) = \mathrm{realize}_n(tx).$$

Theorem 5.6 (All-dimensional constructive coherence) The explicit higher λ-conversion tower $\Sigma_\bullet$ admits an all-dimensional coherence package from the recursive completion of the λ-ω-groupoid core $\mathscr{G}_\lambda$. In particular:

(i) For every $n \in \mathbb{N}$, there is a realization map

$$\mathrm{realize}_n : \mathscr{R}_n(\mathscr{G}_\lambda) \to \Sigma_n.$$

(ii) These realization maps commute strictly with source and target maps.

(iii) Hence the higher λ-conversions form an all-dimensional constructive globular coherence package extending the low-dimensional λ-ω-groupoid core.

Proof We divide the construction into a low-dimensional core and a recursive continuation.

Step 1: dimensions 0–3. At the first four levels there is nothing to prove beyond unpacking the definitions. The objects of $\mathscr{G}_\lambda$ are λ-terms, its 1-cells are explicit reduction zigzags, its 2-cells are explicit homotopies between parallel zigzags, and its 3-cells are explicit homotopies between parallel 2-cells. Thus

$$\mathrm{realize}_0, \mathrm{realize}_1, \mathrm{realize}_2, \mathrm{realize}_3$$

are simply the identity identifications between the recursively completed tower and the explicit tower.

Step 2: dimensions 4–6. Here one must compare two presentations of the same information. On the recursive side, a 4-cell consists of the full boundary data

$$(M, N, p, q, \alpha, \beta, \eta, \theta, \omega),$$

where η, θ are parallel 3-cells and ω is a 4-cell between them in the low-dimensional core. On the explicit λ-tower, by contrast, a 4-cell is presented as a pair of 3-cells together with a higher derivation between those 3-cells. The realization map sends the above tuple to the explicit pair of packaged 3-cells determined by η and θ, together with the image of ω under the functorial action of HigherDeriv on the embedding of 3-cells into the explicit tower.

Exactly the same pattern works in dimensions 5 and 6: one first packages a core 4-cell as an explicit 4-cell, then transports a higher derivation between such cells along that packaging map; similarly for 5-cells. In other words, dimensions $4, 5, 6$ are handled by iterating the same elementary principle:

> *to realize a higher derivation, realize its endpoints first and then map the derivation constructors* $(\mathsf{refl}, \mathsf{symm}, \mathsf{trans})$ *across that endpoint map.*

This produces concrete realization maps $\mathrm{realize}_4, \mathrm{realize}_5, \mathrm{realize}_6$.

Step 3: recursive continuation above dimension 6. Suppose now that $\mathrm{realize}_n$ has already been defined for some $n \geq 6$. An $(n+1)$-cell of the recursive completion is, by definition, a triple (x, y, h) with $x, y \in \mathscr{R}_n(\mathscr{G}_\lambda)$ and $h \in \mathrm{HigherDeriv}(x, y)$. We define

$$\mathrm{realize}_{n+1}(x, y, h) := \big(\mathrm{realize}_n(x),\ \mathrm{realize}_n(y),\ \mathrm{HigherDeriv}(\mathrm{realize}_n)(h)\big).$$

Here $\mathrm{HigherDeriv}(\mathrm{realize}_n)(h)$ denotes the image of h under the map obtained by recursion on the constructors of HigherDeriv:

$$\mathsf{refl}(x) \mapsto \mathsf{refl}(\mathrm{realize}_n(x)), \qquad \mathsf{symm}(h) \mapsto \mathsf{symm}(\mathrm{HigherDeriv}(\mathrm{realize}_n)(h)),$$

$$\mathsf{trans}(h_1, h_2) \mapsto \mathsf{trans}(\mathrm{HigherDeriv}(\mathrm{realize}_n)(h_1), \mathrm{HigherDeriv}(\mathrm{realize}_n)(h_2)).$$

Thus the realization in dimension $n+1$ is forced by the realization in dimension n.

The conceptual point is worth highlighting: no new coherence operations, packaging maps, or case analyses enter the construction above dimension 6. The reason is structural. Below dimension 4, the explicit tower is defined by a bespoke inductive type carrying whiskering, interchange, pentagon, and triangle constructors. In dimensions 4–6, one must translate between that bespoke presentation and the generic recursive completion, which requires the packaging maps of Lemma 5.4. But from dimension 7 onward, *both* towers are defined by the same rule: a cell is a higher derivation between parallel cells of the preceding dimension. The recursive continuation is therefore definitionally an isomorphism of globular sets at every stage, and the realization map is simply the identity modulo the inductive hypothesis. In particular, no finite list of "exceptional dimensions" grows as one climbs the tower: the base and packaging phases are completed once and for all in dimensions 0–6, and the rest is pure induction.

Step 4: compatibility with source and target. For dimensions 0–6, compatibility is checked directly from the formulas: both the recursive tower and the explicit λ-tower compute source by forgetting the last boundary component, and compute target by forgetting the corresponding source component.

For dimensions above 6, let $x = (u, v, h)$ be an $(n+1)$-cell. Then, by definition of the recursive source and target maps,

$$s(x) = u, \qquad t(x) = v.$$

Applying the recursive formula for realization gives

$$\mathrm{realize}_{n+1}(x) = \big(\mathrm{realize}_n(u), \mathrm{realize}_n(v), \mathrm{HigherDeriv}(\mathrm{realize}_n)(h)\big).$$

Hence its source in the explicit tower is exactly $\mathrm{realize}_n(u)$ and its target is exactly $\mathrm{realize}_n(v)$, so

$$s(\mathrm{realize}_{n+1}(x)) = \mathrm{realize}_n(sx), \qquad t(\mathrm{realize}_{n+1}(x)) = \mathrm{realize}_n(tx).$$

This proves strict source and target compatibility in all dimensions.

Step 5: conclusion. The realization maps identify the recursively completed tower of the λ-ω-groupoid core with the explicit higher λ-conversion tower at every dimension. Since the recursive completion is globular by construction and the realization preserves boundaries strictly, the explicit tower inherits the all-dimensional coherence package of the theorem statement. □

5.3. Truncation and the classical theory

An immediate corollary of Theorem 5.6 is that the all-dimensional tower extends, rather than replaces, the ordinary theory of $\beta\eta$-equality.

Corollary 5.7 (0-truncation) The 0-truncation of the all-dimensional ω-groupoid on λ-terms recovers the classical $\beta\eta$-equality:

$$\pi_0(\mathscr{G}_\lambda)(M, N) \iff M =_{\beta\eta} N.$$

Proof The proposition $\pi_0(\mathscr{G}_\lambda)(M, N)$ asserts only that there exists a 1-cell from M to N, that is, an explicit reduction sequence $p : \mathrm{RedSeq}(M, N)$. Forgetting the witness p yields an ordinary $\beta\eta$-conversion from M to N. Conversely, every ordinary $\beta\eta$-conversion admits an explicit reduction-sequence representative. Thus existence of a 1-cell in the all-dimensional tower is equivalent to the classical proposition-level equality relation. □

6. Minimal Semantic Coherence Data

6.1. Three semantic regimes

The semantic development separates three logically distinct layers.

(1) **Bare semantics.** An extensional Kan complex suffices for the interpretation of terms, β- and η-soundness, the interpretation of explicit reduction sequences, and the semantic images of the explicit 2- and 3-cells already present in the syntax.

(2) **Coherent semantics.** A coherent extensional Kan complex adds explicit 3-dimensional comparison data for associativity and whiskering. This makes it possible to compare different semantic parenthesizations of composite interpreted paths.

(3) **Strict semantics.** In a strict extensional Kan complex, the required coherent comparison data are derived axiom-freely from uniqueness of horn fillers, so no additional chosen coherence fields are needed beyond the extensional λ-model structure.

We now show that a smaller front-facing seed already generates the full associator and pentagon package needed later.

6.2. The reduced front-seed boundary

Definition 6.1 (Front-seed coherent extensional Kan complex) A *front-seed coherent extensional Kan complex* is an extensional Kan complex K equipped with the following additional 3-cell data.

(FS1) **WLWR comparison.** For every composable 1-cells

$$a \xrightarrow{\alpha} b \xrightarrow{\beta,\gamma} c \xrightarrow{\delta} d$$

and every 2-cell $\eta : \beta \Rightarrow \gamma$, a chosen 3-cell

$$\mathrm{wr}(\mathrm{wl}(\alpha, \eta), \delta) \implies \mathrm{ass}(\alpha, \beta, \delta) \cdot \mathrm{wl}(\alpha, \mathrm{wr}(\eta, \delta)) \cdot \mathrm{ass}(\alpha, \gamma, \delta)^{-1}.$$

In words: right-whiskering a left-whiskered 2-cell agrees, up to the appropriate associator corrections, with first right-whiskering and then left-whiskering.

(FS2) **Inner-right-front pentagon contraction.** For every composable quadruple p, q, r, s, a chosen 3-cell contracting the inner-right-front face of the semantic pentagon to the reflexive 2-cell on the edge labeled by $q \cdot (r \cdot s)$ before left-whiskering by p.

The second datum is deliberately asymmetric: it does not assert the entire pentagon at once, but only contracts one distinguished front face. The theorem below says that, together with WLWR, this already forces the higher coherence needed for associator comparison and for the full pentagon at interpreted reduction sequences.

It is useful to describe the geometric content of the two seeds more precisely.

FS1 (WLWR comparison): front-face horn filling for mixed whiskering. Given a 2-cell $\eta : \beta \Rightarrow \gamma$ flanked by 1-cells α on the left and δ on the right, there are two natural ways to extend η to the full composite boundary: first left-whisker by α and then right-whisker by δ, or first right-whisker by δ and then left-whisker by α. These two expressions land at the same pair of endpoints $\alpha \cdot \beta \cdot \delta$ and $\alpha \cdot \gamma \cdot \delta$, but they are not *a priori* the same 2-cell because the whiskering operations interact with the associativity of path composition. Seed (FS1) asserts that the discrepancy is controlled by the associator: the two mixed whiskerings differ by exactly the pair of associator 2-cells that rebrackets the composite. In simplicial terms, FS1 fills

a specific 3-horn whose two front faces are the two mixed-whiskering composites and whose remaining faces are associator instances.

FS2 (inner-right-front pentagon contraction): partial boundary seeding. The pentagon coherence for four composable 1-cells p, q, r, s is a 3-cell whose boundary consists of five associator 2-cells arranged around the boundary of a 4-simplex. Rather than positing the full pentagon 3-cell as primitive, FS2 provides only the contraction of the *inner-right-front* face—the face built from the associator $\mathrm{ass}(q, r, s)$ whiskered by p on the left. This contracts a particular sub-composite of the pentagon boundary to the identity 2-cell on the edge labeled by $q \cdot (r \cdot s)$ before left-whiskering by p. Geometrically, FS2 is a partial horn-filling datum: it specifies one non-degenerate face of the pentagon 4-horn, from which the remaining faces can be assembled using the WLWR comparison of FS1 together with the standard Kan-complex horn-filling machinery.

The significance of Theorem 6.2 is a precise reduction of the semantic coherence package actually used later. In the formal development, the coherent semantic interface is already separated into two chosen ingredients: a WLWR comparison and a full pentagon witness. The front-seed theorem sharpens the second ingredient: the full pentagon field may be replaced by the smaller inner-right-front contraction, from which one derives the recursive associator comparison, the semantic pentagon on interpreted reduction sequences, and the source/target/shell bridge theorems for the explicit syntactic pentagon. Thus the gain is not an appeal to an unspecified larger coherence package, but a concrete reduction from a full pentagon datum to a smaller front-facing seed, while keeping WLWR explicit.

Theorem 6.2 (Reduced front-seed coherence) Let K be a front-seed coherent extensional Kan complex. Then:

(i) For any reduction sequences $p_1, \ldots, p_k$, every interpreted parenthesization of their composite is related to the standard left-associated composite by a chain of semantic associators and inverses.
(ii) The semantic pentagon identity holds for interpreted reduction sequences.
(iii) The explicit syntactic pentagon witnesses admit source, target, and shell bridge theorems to the structural semantic pentagon shells.

Proof We prove the three claims in order.

Step 1: from WLWR to the recursive associator comparison. The basic inductive problem is the following. Let p, q, r be explicit reduction sequences. There is a *structural associator shell* comparing the two semantic composites

$$((p \cdot q) \cdot r) \qquad \text{and} \qquad p \cdot (q \cdot r),$$

obtained by interpreting explicit paths and then composing them in K. There is also an *equality-generated* comparison coming from the literal associativity equality of concatenation of reduction sequences. One must show that these two semantic 2-cells are themselves connected by a semantic 3-cell.

For a composite whose leftmost factor is a single generating step, the only non-formal obstruction is a nested whiskering term of the shape

$$\mathrm{wr}(\mathrm{wl}(\alpha, \eta), \delta),$$

where $\eta : \beta \Rightarrow \gamma$, α is the semantic image of the head step, and δ is the semantic image of r. Seed (FS1) replaces exactly this nested whisker by the normalized expression

$$\mathrm{ass}(\alpha, \beta, \delta) \ \cdot \ \mathrm{wl}(\alpha, \mathrm{wr}(\eta, \delta)) \ \cdot \ \mathrm{ass}(\alpha, \gamma, \delta)^{-1}.$$

After this normalization, the head-step case matches the standard associator shell and closes.

One then argues by induction on the first reduction sequence p. If $p = \mathsf{refl}$, the comparison is immediate. If p begins with a forward or backward step, write p as a head step followed by a shorter tail. The WLWR comparison handles the head-step normalization just described, and the induction hypothesis handles the tail. Vertical composition of these 3-cells produces a 3-cell from the structural associator shell of (p, q, r) to the equality-generated associator. Iterating this argument gives the comparison for arbitrary parenthesized composites of any finite list $p_1, \ldots, p_k$: one changes parenthesization one associator at a time, and each elementary move is controlled by the preceding inductive comparison.

Step 2: from the front pentagon seed to the full semantic pentagon. Now fix four reduction sequences p, q, r, s. The semantic pentagon compares the two composites

$$\mathrm{ass}(p \cdot q, r, s) \ \cdot \ \mathrm{ass}(p, q, r \cdot s)$$

and

$$(\mathrm{ass}(p, q, r) * s) \ \cdot \ \mathrm{ass}(p, q \cdot r, s) \ \cdot \ (p * \mathrm{ass}(q, r, s)),$$

where $*$ denotes whiskering. Seed (FS2) gives a contraction of the inner-right-front face of the pentagon. This is not yet the full pentagon, but it determines one crucial missing face of the corresponding 4-dimensional horn. The WLWR comparison

from Step 1 supplies the whiskering rebracketing needed to identify the remaining front-facing pieces with the appropriate associator composites. Once these auxiliary faces are in place, the horn has all of its boundary fixed except for the desired pentagon face, so the Kan filling machinery provides the missing 3-cell. Equivalently, the front seed canonically upgrades K to a coherent extensional Kan complex, and in that upgraded structure the usual semantic pentagon is available. Evaluating it on the interpreted sequences p, q, r, s yields the required direct semantic pentagon for interpreted reduction sequences.

Step 3: bridge theorems from syntax to semantic shells. The explicit syntactic higher calculus already contains an associator 2-cell and a pentagon 3-cell between explicit reduction sequences. These objects can be interpreted in any bare extensional Kan complex, so in particular in our front-seed model K. However, their semantic images land a priori in the "syntactic" presentation of the boundary, while Step 1 and Step 2 produce the "structural" semantic shells built from the Kan-complex associator and whiskering operations.

The associator bridge from Step 1 identifies the semantic image of the explicit syntactic associator with the structural associator shell. Applying this bridge to the two associators on the source side of the pentagon yields the *source bridge*; applying it to the three associator pieces on the target side yields the *target bridge*. Finally, compose

(a) the source bridge,
(b) the interpreted syntactic pentagon 3-cell, and
(c) the inverse of the target bridge.

The result is a 3-cell from the structural source shell to the structural target shell. This is exactly the shell bridge theorem. Thus the explicit syntactic pentagon and the structural semantic pentagon are not merely parallel facts: they are connected by explicit comparison 3-cells.

This proves (i), (ii), and (iii). □

Remark 6.3 (Open strengthening target) The remaining conceptual question is whether the two front-seed data can be derived from the bare extensional Kan interface alone. A positive answer would show that no additional semantic coherence fields are needed even for the associator and pentagon package. At present, what is proved is sharper but more modest: the full coherent interface is unnecessary, and the minimal boundary currently known to suffice is precisely the WLWR comparison together with the inner-right-front pentagon contraction.

7. The K_∞ Model

The K_∞ model was introduced in Paper II [9], which defined the inverse-limit construction, proved algebraicity and bounded completeness (Proposition 4.1), and exhibited the separation phenomenon of Example 8.1. What was not established there—and what the present section provides—is the *exact reflexive packaging*: explicit, globally continuous reify and reflect maps together with a stagewise application formula that pins down the relationship between the finite stages and the limit. The packaging theorem (Theorem 7.6) also proves a retract law strictly stronger than the standard domain-theoretic requirement $h(k(x)) \leq x$, showing that $h \circ k = \mathrm{id}$ holds on the nose.

7.1. Construction

The object K_∞ is obtained as the inverse limit of the reflexive tower

$$K_0,\ K_1,\ K_2,\ \ldots, \qquad K_0 = \mathbb{N}^+, \qquad K_{n+1} = [K_n \to K_n].$$

At each stage there is a projection pair

$$f_n^+ : K_n \longrightarrow K_{n+1}, \qquad f_n^- : K_{n+1} \longrightarrow K_n$$

with

$$f_n^- \circ f_n^+ = \mathrm{id}_{K_n}, \qquad f_n^+\big(f_n^-(u)\big) \leq u \quad (u \in K_{n+1}).$$

Thus every successor stage is a reflexive enlargement of the previous one.

Definition 7.1 (The tower $K_0, K_1, \ldots$) Set $K_0 = \mathbb{N}^+$, the flat domain on the non-trivial homotopy-group witness set together with bottom, and define inductively

$$K_{n+1} = [K_n \to K_n].$$

The transition maps are the projection pairs (f_n^+, f_n^-) above.

Definition 7.2 (K_∞) The inverse limit K_∞ is the set of coherent threads

$$K_\infty = \varprojlim_n K_n = \Big\{(x_n)_{n\geq 0} \in \prod_{n\geq 0} K_n : f_n^-(x_{n+1}) \simeq x_n \text{ for all } n\Big\},$$

ordered coordinatewise. We write

$$\pi_n : K_\infty \to K_n$$

for the nth projection.

The domain-theoretic content of the construction is the following.

Proposition 7.3 (Proposition 4.1) K_∞ is a homotopy Scott domain: it is algebraic and bounded complete.

Proof We first analyze the finite stages.

Bounded completeness of the stages. The base $K_0 = \mathbb{N}^+$ is a flat domain, hence bounded complete. If K_n is bounded complete, then the function space $K_{n+1} = [K_n \to K_n]$ is bounded complete under the pointwise order, since least upper bounds of upper-bounded families are computed pointwise. Therefore every K_n is bounded complete.

Algebraicity of the stages. The base stage K_0 is algebraic: every non-bottom element is compact, and every element is the least upper bound of the compact elements below it. Assume now that K_n is algebraic. We show that $K_{n+1} = [K_n \to K_n]$ is algebraic.

For $a, b \in K_n$ with a compact and b compact-below $f(a)$, let $[a \Rightarrow b]$ denote the usual compact step map which sends arguments above a to b and arguments below a to bottom. Finite joins of such step maps are still compact. Let $\mathcal{A}_f$ be the set of all finite joins of compact step maps lying below f. Then $\mathcal{A}_f$ is directed under inclusion of the underlying finite families, and every member of $\mathcal{A}_f$ is compact-below f.

To see that f is the least upper bound of $\mathcal{A}_f$, fix $x \in K_n$. Since K_n is algebraic, the value $f(x)$ is the least upper bound of the compact elements below it. Each such compact element is already captured by some step approximation built from a compact approximation to x. Hence evaluation at x of the directed family $\mathcal{A}_f$ is cofinal in the compact-below approximants to $f(x)$, so its pointwise supremum is exactly $f(x)$. Since this holds for every x, $f = \sup \mathcal{A}_f$, and K_{n+1} is algebraic.

Thus every finite stage K_n is an algebraic bounded-complete domain.

Passage to the inverse limit. For $x \in K_\infty$, define its nth finite-stage approximation by

$$a_n(x) := f_{n,\infty}(\pi_n(x)),$$

where $f_{n,\infty} : K_n \to K_\infty$ is the canonical exact embedding of the nth stage into the limit. By construction,

$$\pi_m(a_n(x)) = \pi_m(x) \qquad (m \leq n),$$

so each $a_n(x)$ agrees with x on the first $n+1$ coordinates; in particular $a_n(x) \leq x$. Moreover the family $(a_n(x))_{n\geq 0}$ is directed, since passing from stage n to stage $n+1$ only refines the approximation.

Now let $c \ll x$ be compact-below in K_∞. Compactness in an inverse limit is finite-stage: there is some stage n and some compact $b \ll \pi_n(x)$ in K_n such that

$$c \leq f_{n,\infty}(b) \leq a_n(x).$$

Hence the directed family $\{a_n(x)\}_{n\geq 0}$ is cofinal among compact approximants below x. Therefore x is a least upper bound of this family, and the chosen supremum of the family is equivalent to x in the induced homotopy order:

$$x \simeq \sup_{n\geq 0} a_n(x).$$

Thus every element of K_∞ is a least upper bound of compact elements below it. This proves algebraicity of K_∞.

Finally, bounded completeness of K_∞ is obtained coordinatewise. If $X \subseteq K_\infty$ is upper-bounded, then each projected family $\pi_n(X) \subseteq K_n$ is upper-bounded and therefore has a least upper bound in K_n. The projection-pair coherence ensures that these coordinatewise suprema again form a coherent thread, which is the least upper bound of X in K_∞. □

7.2. Exact reflexive packaging

The central structural fact is that the inverse limit is itself reflexive. In fact, the projection-pair machinery on the tower produces global reify and reflect maps. The proof of this fact assembles three technical ingredients that are implicit in the inverse-limit construction but are worth isolating as standalone statements, both because they clarify the argument and because they are reused in the proof-relevant witness analysis of Section 8.

Lemma 7.4 (Density of finite-stage approximants) For every $x \in K_\infty$, define the *nth finite-stage approximant*

$$a_n(x) := f_{n,\infty}(\pi_n(x)) \in K_\infty.$$

Then:

(i) $a_n(x) \leq x$ for every n,
(ii) the family $(a_n(x))_{n\geq 0}$ is directed, and
(iii) $x \simeq \sup_{n\geq 0} a_n(x)$.

In particular, K_∞ is generated, up to chosen-supremum equivalence, by the embedded finite stages $\bigcup_n f_{n,\infty}(K_n)$.

Proof Clause (i) holds because $a_n(x)$ agrees with x on coordinates 0 through n (by the retract property $\pi_n \circ f_{n,\infty} = \mathrm{id}$) and is $\leq x$ on all higher coordinates (by iterated application of the section inequality $f_m^+ f_m^- \leq \mathrm{id}$). Clause (ii) follows because passing from stage n to $n+1$ only refines the approximation: $a_n(x) \leq a_{n+1}(x)$ by the compatibility of the projection pairs. Clause (iii) was established in the proof of Proposition 4.1 (algebraicity of K_∞): every compact element below x is already below some $a_n(x)$, so the directed family is cofinal among the compact approximants to x, hence has x as a least upper bound; the chosen supremum is therefore equivalent to x. □

Lemma 7.5 (Coherence of stage restrictions) Let $g : K_\infty \to K_\infty$ be a continuous endomap and define its *stage-n restriction* by

$$r_n(g) := \pi_n \circ g \circ f_{n,\infty} \ \in \ [K_n \to K_n] = K_{n+1}.$$

Then the family $(r_n(g))_{n\geq 0}$ satisfies the coherence condition

$$f_n^-\big(r_{n+1}(g)\big) \simeq r_n(g) \qquad \text{for all } n \geq 0.$$

In particular, the shifted thread $\big(f_0^-(r_0(g)),\, r_0(g),\, r_1(g),\, \ldots\big)$ is a well-defined element of K_∞.

Proof Fix $a \in K_n$. Applying f_n^- to the map $r_{n+1}(g)$ and evaluating at a gives

$$f_n^-\big(r_{n+1}(g)\big)(a) = \pi_n\big(g(f_{n+1,\infty}(f_n^+(a)))\big).$$

Since $f_{n+1,\infty} \circ f_n^+$ and $f_{n,\infty}$ produce the same coherent thread in K_∞ (both agree on all coordinates up to n and satisfy the tower coherence above n), continuity of g gives $g(f_{n+1,\infty}(f_n^+(a))) = g(f_{n,\infty}(a))$. Projecting to level n yields $r_n(g)(a)$. Since a was arbitrary, $f_n^-(r_{n+1}(g)) = r_n(g)$ as required. The "in particular" clause follows because this coherence condition is exactly the defining property of a thread in the inverse limit K_∞. □

These two lemmas decompose the proof of the following theorem into identifiable modules: Lemma 7.4 provides the density argument that drives the section law, and Lemma 7.5 provides the well-definedness of the reify map.

Theorem 7.6 (Exact K_∞ reflexive packaging) There exist globally continuous maps

$$h : [K_\infty \to K_\infty] \to K_\infty, \qquad k : K_\infty \to [K_\infty \to K_\infty]$$

such that

$$k \circ h = \mathrm{id}_{[K_\infty \to K_\infty]}, \qquad h \circ k = \mathrm{id}_{K_\infty}.$$

The induced application operation

$$\mathsf{app}(x, y) := k(x)(y)$$

is jointly continuous. If $f_{n,\infty} : K_n \to K_\infty$ denotes the canonical stage embedding, then for every $x \in K_\infty$, every $y \in K_n$, and every $n \geq 0$,

$$\pi_n\big(\mathsf{app}(x, f_{n,\infty}(y))\big) = \pi_{n+1}(x)(y).$$

Proof We construct h and k directly from the finite stages.

Step 1: exact finite-stage embeddings. For each n there is a canonical map

$$f_{n,\infty} : K_n \to K_\infty$$

obtained as follows: below level n one repeatedly applies the projections f_i^-, at level n one keeps the given element, and above level n one repeatedly applies the embeddings f_i^+. Explicitly, the thread $f_{n,\infty}(u)$ is the unique coherent sequence whose nth

coordinate is u, whose lower coordinates are its iterated projections, and whose higher coordinates are its iterated embeddings. By construction,

$$\pi_n \circ f_{n,\infty} = \mathrm{id}_{K_n},$$

and successive embeddings are compatible with the tower:

$$f_{n+1,\infty} \circ f_n^+ \simeq f_{n,\infty}.$$

Step 2: finite-stage restrictions of endomaps. Given a continuous endomap $g : K_\infty \to K_\infty$, define its stage-$n$ restriction by

$$r_n(g) := \pi_n \circ g \circ f_{n,\infty} \in [K_n \to K_n] = K_{n+1}.$$

These restrictions are compatible:

$$f_n^-\big(r_{n+1}(g)\big) \simeq r_n(g).$$

Indeed, for $a \in K_n$,

$$f_n^-\big(r_{n+1}(g)\big)(a) = \pi_n\big(g(f_{n+1,\infty}(f_n^+(a)))\big) \simeq \pi_n\big(g(f_{n,\infty}(a))\big) = r_n(g)(a),$$

because $f_{n+1,\infty} \circ f_n^+$ and $f_{n,\infty}$ define the same thread in the inverse limit.

The coherence condition of Lemma 7.5 guarantees that we may assemble the coherent thread

$$h(g)_0 := f_0^-(r_0(g)), \qquad h(g)_{n+1} := r_n(g) \quad (n \geq 0).$$

This defines a map

$$h : [K_\infty \to K_\infty] \to K_\infty,$$

and continuity is immediate because each coordinate $h(g)_{n+1} = r_n(g)$ is obtained by composing continuous maps.

Step 3: finite-stage application shadows. For $x, y \in K_\infty$, define the nth binary approximation to application by

$$k_n(x, y) := f_{n,\infty}\big(\pi_{n+1}(x)(\pi_n(y))\big) \in K_\infty.$$

Each k_n is jointly continuous. Moreover the family $(k_n)_{n\geq 0}$ is monotone:

$$k_n(x, y) \leq k_{n+1}(x, y).$$

This follows from two facts: first, the thread coherence of x identifies $\pi_n(x)$ with the f_n^--projection of $\pi_{n+1}(x)$; second, the section inequality $f_n^+ f_n^- \leq \mathrm{id}$ implies that evaluating at the lifted argument in stage $n+1$ dominates the stage-n value. Hence the directed supremum

$$k(x, y) := \sup_{n\geq 0} k_n(x, y)$$

exists in K_∞ and is jointly continuous. Currying gives

$$k : K_\infty \to [K_\infty \to K_\infty].$$

Step 4: the stagewise formula. Fix $x \in K_\infty$ and $y \in K_n$. Restrict the map $k(x)$ along the exact embedding $f_{n,\infty}$. On the one hand, the nth approximant $k_n(x, f_{n,\infty}(y))$ has nth coordinate exactly $\pi_{n+1}(x)(y)$. On the other hand, every higher approximant has nth coordinate below the same value, so after projecting to level n the entire directed family is bounded above by $\pi_{n+1}(x)(y)$ and contains an element attaining that upper bound. Since projection preserves directed suprema,

$$\pi_n\big(k(x)(f_{n,\infty}(y))\big) = \pi_{n+1}(x)(y).$$

This is exactly the stated stagewise application formula.

Step 5: the section law $k \circ h = \mathrm{id}$. Let $g : K_\infty \to K_\infty$. For every stage n and every $y \in K_n$, the previous formula applied to $h(g)$ gives

$$\pi_n\big(k(h(g))(f_{n,\infty}(y))\big) = \pi_{n+1}(h(g))(y) = r_n(g)(y) = \pi_n\big(g(f_{n,\infty}(y))\big).$$

Thus $k(h(g))$ and g agree on every embedded finite stage.

Now fix $z \in K_\infty$ and use the density of finite-stage approximants (Lemma 7.4): the directed family of finite-stage approximants of z has z as a least upper bound, and its chosen supremum is equivalent to z. Since both $k(h(g))$ and g are continuous and agree on every member of this generating directed family, they agree on the chosen supremum. The equivalence with z then transports this equality back to z, and antisymmetry of the stage orders forces $k(h(g))(z) = g(z)$. Hence $k(h(g)) = g$.

Step 6: the retract law $h \circ k = \mathrm{id}_{K_\infty}$. For $x \in K_\infty$ and $n \geq 0$,

$$\pi_{n+1}(h(k(x))) = r_n(k(x)).$$

Evaluating this map at $y \in K_n$ and using the stagewise formula just proved, we obtain

$$\pi_{n+1}(h(k(x)))(y) = \pi_n\big(k(x)(f_{n,\infty}(y))\big) = \pi_{n+1}(x)(y).$$

Therefore the $(n+1)$st coordinates of $h(k(x))$ and x coincide for all $n \geq 0$. At level 0 one gets

$$\pi_0(h(k(x))) = f_0^-(\pi_1(x)) = \pi_0(x)$$

by thread coherence. Hence in fact

$$h(k(x)) = x.$$

Finally, $\mathsf{app}(x, y) = k(x)(y)$ is jointly continuous because it is the composite of the continuous map

$$(x, y) \longmapsto (k(x), y)$$

with the continuous evaluation map on the function space $[K_\infty \to K_\infty]$. □

The maps h and k are implemented in the Lean formalization by projection-pair data at each finite stage. The following code fragment illustrates that interface; Section 8 then uses the stagewise formula of Theorem 7.6 to separate proof-relevant witness classes inside K_∞.

Listing 1. Projection pairs for the K_n tower (from `KInfinity/Construction.lean`).

```
structure ProjectionPair
    (A : CompleteHomotopyPartialOrder)
    (B : CompleteHomotopyPartialOrder) where
  emb : ContinuousMap A B
  proj : ContinuousMap B A
  retract : ContinuousMap.comp proj emb = ContinuousMap.id A
  section_le :
    forall x : B.Obj, B.Rel (emb (proj x)) x
```

8. Proof-Relevant Witness Separation

Paper II [9] established, via the witness-separation span restated below as Example 8.1, that the K_∞ model is nontrivial: two distinct contraction patterns from the same redex receive semantically distinct images. That result, however, was a point-level observation—it did not analyze the structure of the reduction witnesses themselves, nor did it trace the separation through the recursively completed higher-cell tower above the explicit core dimensions.

The present section has two logically distinct parts. First, we introduce a *witness language* that remembers which contraction was chosen and classifies every explicit one-step witness by a canonical β or η tag. This is the model-specific input. Second, we show that once the resulting point-level dichotomy is established, the absence of connecting 1-cells and higher cells is forced by the equality-generated structure of the K_∞ higher-cell tower, including the recursively completed dimensions.

8.1. A basic witness-separation example

Paper II [9] proved the non-triviality of K_∞ by means of the following concrete span, which we restate here for direct reference.

Example 8.1 (Witness-separation span) Consider the source

$$M = ((\lambda z.\, xz)\, y)$$

and the common target

$$N = xy.$$

The two basic one-step contractions are displayed explicitly as

$$((\lambda z.\, xz)\, y) \;\to_\beta\; xz[y/z] = xy$$

and

$$((\lambda z.\, xz)\, y) \;\to_\eta\; xy,$$

where in the second line the η-contraction is performed in the function part $\lambda z.\, xz \to_\eta x$. Thus the same source and target admit two distinguished one-step witnesses: the direct β-contraction and the direct η-contraction through the function part.

The point of the proof-relevant refinement is that we do not merely remember that the endpoints are propositionally $\beta\eta$-equal; we remember *which contraction witness was chosen.* A general proof-relevant explicit 1-witness from M to N may contain reflexive padding before or after the essential contraction, but within this witness-separation span the witness language still falls into exactly two reachability classes: the β-class and the η-class. Thus every proof-relevant 1-witness on this span carries a canonical tag

$$\mathrm{tag}(t) \in \{\beta, \eta\}.$$

Figure 2 records this fixed span and the two canonical semantic images attached to its witness classes.

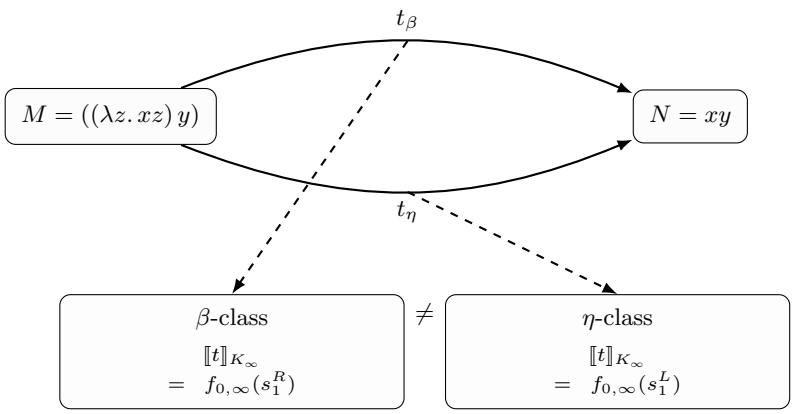


Figure 2 The witness-separation span of Example 8.1. The same source and target admit a β-witness and an η-witness, but their canonical K_∞ interpretations land at distinct points.

8.2. Witness languages

Definition 8.2 (Explicit witnesses) A β_1-*witness* for terms (M, N) is an explicit one-step β-reduction $M \to_\beta N$ together with the reduction-sequence datum it generates. An η_1-*witness* is defined analogously. More generally, a 1-*term witness* is an explicit reduction witness obtained from these generators by composition and reflexive padding.

In this section we restrict attention to the fixed witness-separation span of Example 8.1

$$M = ((\lambda z.\, xz)\, y), \qquad N = xy.$$

To each explicit 1-term witness t from M to N we attach two pieces of semantic data:

1. a continuous equality ϵ_t witnessing that the source and target terms have the same ordinary denotation in the reflexive semantics, and
2. a distinguished point of K_∞ recording whether the witness belongs to the β-class or to the η-class.

The second piece is what detects proof relevance. Let s_1^R and s_1^L be the right and left poles in the S^1-summand of the non-trivial base object used to define $K_0 = \mathbb{N}^+$. If $\mathrm{tag}(t) = \beta$, we set

$$[\![t]\!]_{K_\infty} := f_{0,\infty}(s_1^R),$$

and if $\mathrm{tag}(t) = \eta$, we set

$$[\![t]\!]_{K_\infty} := f_{0,\infty}(s_1^L).$$

Thus the interpretation depends only on the essential witness class, not on the amount of reflexive padding surrounding it. We now make this classification and its stability precise before stating the main separation theorem.

Lemma 8.3 (Witness classification on the witness-separation span) Let t be an explicit 1-term witness from the source $M = ((\lambda z.\, xz)\, y)$ to the target $N = xy$. Then exactly one of the following holds:

(a) t is obtained from the direct β-contraction by inserting reflexive witnesses on the left and/or right;
(b) t is obtained from the direct η-contraction by inserting reflexive witnesses on the left and/or right.

In particular, the canonical tag $\mathrm{tag}(t) \in \{\beta, \eta\}$ is well-defined.

Proof Proceed by structural induction on the explicit witness language. The base constructors are the direct β- and η-witnesses together with reflexivity. The reflexive case cannot occur between the chosen source and target because $M \neq N$, so any witness from M to N must be classified by one of the two nontrivial contractions.

For a composite witness, once a nontrivial β- or η-witness has appeared, the remaining factor can contribute only reflexive padding. Indeed, the witness-separation span admits no third nontrivial contraction pattern: the source has exactly the direct outer β-contraction and the inner η-contraction through the function part, and the target xy supports no further forward β- or η-step. Hence every sequential composite factors as one distinguished nontrivial contraction together with reflexive witnesses on the left and/or right.

The two classes are disjoint because the direct β- and η-steps are syntactically distinct and have different chosen semantic tags. Therefore exactly one of (a) or (b) holds, and the tag is well-defined. □

Corollary 8.4 (Stability under reflexive padding) Let t and t' be explicit 1-term witnesses from M to N that differ only by insertion or removal of reflexive subpaths. Then $\mathrm{tag}(t) = \mathrm{tag}(t')$, and hence $[\![t]\!]_{K_\infty} = [\![t']\!]_{K_\infty}$.

Proof By Lemma 8.3, the two classes are defined precisely by adjoining reflexive witnesses to the direct β- or η-contraction. Inserting or removing reflexive padding therefore cannot change whether a witness is of β-type or η-type. Hence $\mathrm{tag}(t) = \mathrm{tag}(t')$, and the K_∞-interpretation was defined in terms of the tag alone. □

With these preliminaries in hand, we can state the main separation theorem.

Theorem 8.5 (Proof-relevant witness separation) Let t_β and t_η be the explicit β_1- and η_1-witnesses from Example 8.1. Their canonical K_∞ interpretations are:

(i) distinct as K_∞-points,
(ii) not connected by any K_∞-path (1-cell),
(iii) not connected by any 2-cell, 3-cell, or 4-cell in the explicit K_∞ higher-cell tower,
(iv) not connected by any n-cell for any $n \geq 5$ in the recursively completed tower.

The same conclusion holds for any pair of explicit 1-witnesses whose canonical tags are β and η, respectively, on this same witness-separation span.

Proof We divide the argument into classification, point-level separation, and transport through the higher dimensions.

1. Classification of witnesses. By Lemma 8.3, every proof-relevant explicit 1-witness t from the source M to the target N of Example 8.1 lies in exactly one of two classes: the β-class, obtained from the direct β-contraction by reflexive padding, or the η-class, obtained analogously from the direct η-contraction. The canonical tag $\mathrm{tag}(t) \in \{\beta, \eta\}$ is therefore well-defined. By Corollary 8.4, the tag—and hence the K_∞-interpretation—is invariant under insertion or removal of reflexive subpaths.

2. Canonical interpretation. The interpretation of t is the pair $(\epsilon_t, [\![t]\!]_{K_\infty})$ consisting of its continuous semantic witness ϵ_t together with the point

$$[\![t]\!]_{K_\infty} = \begin{cases} f_{0,\infty}(s_1^R), & \text{if } \mathrm{tag}(t) = \beta, \\ f_{0,\infty}(s_1^L), & \text{if } \mathrm{tag}(t) = \eta. \end{cases}$$

In particular,

$$\pi_0([\![t_\beta]\!]_{K_\infty}) = s_1^R, \qquad \pi_0([\![t_\eta]\!]_{K_\infty}) = s_1^L.$$

3. Separation at dimension 0. Since s_1^R and s_1^L are distinct poles of the chosen S^1-summand, they are distinct already in K_0. Hence

$$[\![t_\beta]\!]_{K_\infty} \neq [\![t_\eta]\!]_{K_\infty},$$

because equality in K_∞ would imply equality of all projections, in particular equality of the 0th projections. This proves (i).

4. No 1-cells. The canonical K_∞-tower used here is the equality-type ω-groupoid on the carrier of K_∞. Consequently a 1-cell from x to y is precisely an identification $x = y$. Since the two points are already distinct by the previous step, there is no 1-cell between them. This proves (ii).

5. Transport through dimensions 2 *to* 4. A 2-cell can exist only between parallel 1-cells; therefore the existence of a 2-cell between the β- and η-interpretations would in particular produce a 1-cell between them, impossible by (ii). The same argument iterates: a 3-cell has as boundary a pair of 2-cells, and a 4-cell has as boundary a pair of 3-cells. Hence emptiness propagates inductively upward through the explicit low-dimensional hierarchy. This proves (iii).

6. Recursive higher dimensions. For dimensions ≥ 5, the recursively completed tower is obtained from the low-dimensional K_∞-cells by iterated identity completion. In that completed tower every cell of dimension $m \geq 5$ has equal 0-source and

0-target. Concretely, this is because the explicit 5- and 6-cells already collapse their object-level boundary, and every higher cell is then generated by repeated identity extension from lower-dimensional data.

Therefore, if there were an $(n+5)$-cell in the recursive tower with 0-source $[\![t_\beta]\!]_{K_\infty}$ and 0-target $[\![t_\eta]\!]_{K_\infty}$, the equality of 0-source and 0-target for all such cells would force

$$[\![t_\beta]\!]_{K_\infty} = [\![t_\eta]\!]_{K_\infty},$$

contradicting the dimension-0 separation already proved. This establishes (iv).

7. Uniformity. Nothing in the preceding argument used any special feature of the two distinguished witnesses beyond their tags on the fixed witness-separation span. Any witness on that span classified as β is interpreted by the same canonical β-point, and any witness classified as η is interpreted by the same canonical η-point. Hence the same separation theorem holds for every pair of witnesses on this span lying in opposite classes. □

Remark 8.6 Theorem 8.5 should be read in two layers. Its genuinely model-specific input is the canonical β/η dichotomy on the witness language of Example 8.1 and the corresponding separation at dimension 0. Once that input is fixed, the higher-dimensional clauses record that the recursive equality-generated tower does not create connecting cells across the two classes. Thus the theorem isolates both the base semantic distinction and the precise sense in which that distinction propagates through the proof-relevant semantics.

9. Discussion and Future Work

9.1. Summary of results

We have presented four theorem packages on the theory of higher λ-models:

(A) All-dimensional constructive coherence for higher λ-conversions (Theorem 5.6).
(B) Minimal semantic coherence data for associator/pentagon reasoning (Theorem 6.2).
(C) Proof-relevant witness separation for K_∞ (Theorem 8.5).
(D) Exact reflexive packaging and application for K_∞ (Theorem 7.6).

These four results have different mathematical roles. Theorem 7.6 is the most concrete model-specific theorem: it gives an exact reflexive structure on K_∞ with explicit stagewise formulas. Theorem 6.2 isolates the minimal coherence interface required to transport associator and pentagon arguments into that setting. Theorems 5.6 and 8.5 are more structural: the former compares the explicit low-dimensional tower with its recursive equality-generated continuation, while the latter shows that the point-level β/η separation on the witness-separation span remains stable under that continuation.

9.2. Comparison with related work

We organize the discussion into three threads: direct antecedents within the higher λ-model programme, the broader higher-categorical and homotopical background, and the classical domain-theoretic lineage.

Direct antecedents. The immediate foundation is the two-paper programme of Martínez-Rivillas and de Queiroz. Paper I [10] introduced the extensional Kan semantics, defined the explicit tower of n-conversions, and proved the main inclusion $\mathrm{TH}_{\lambda=} \subseteq$ HoTFT. Paper II [9] constructed K_∞ and exhibited the separation restated here as Example 8.1. The present work begins from those results but addresses four specific remaining gaps. For Theorem 5.6, the missing point is not the existence of recursive higher-derivation syntax, but the comparison between the explicit low-dimensional tower and the equality-generated continuation together with the proof of strict globular compatibility for that continuation. For Theorem 6.2, the missing point is the isolation of the smaller WLWR-plus-front-seed interface sufficient for the later semantic arguments. For Theorem 8.5, Paper II gives only point-level separation; what was missing was a witness-level classification on the witness-separation span together with an account of how that separation behaves in the recursive tower. For Theorem 7.6, the missing point is the exact reflexive packaging of K_∞ by globally continuous reify, reflect, and application maps with explicit stagewise formulas and a strict retract law.

Earlier in the same programme, Martínez-Rivillas and de Queiroz [8] developed a homotopy domain theory that enriches the classical Scott semantics with an explicit homotopical layer, and in [11] they studied the ∞-groupoid generated by an arbitrary topological λ-model. Both papers anticipate the idea that the conversion structure of the λ-calculus has a nontrivial higher-categorical life, but neither constructs the all-dimensional coherence package or the witness-level separation that the present paper provides.

A closely related syntactic thread is the computational-paths programme of de Veras, Ramos, de Queiroz, and de Oliveira [3], which showed that the paths generated by directed reduction steps in a rewriting system form a weak groupoid. Their construction provides a syntactic template for the 1-dimensional layer of our tower: the reduction sequences between λ-terms are precisely computational paths, and the explicit 2-cells (homotopies between reduction sequences) can be viewed as the coherence data that upgrade the resulting path groupoid to a 2-groupoid. The present paper extends that picture to all dimensions by showing that the recursive completion of the low-dimensional core yields a genuine ω-groupoid (Theorem 5.6).

Higher-categorical and homotopical background. The coherence machinery that underlies Theorem 5.6 has a long history in higher category theory. Batanin [2] introduced monoidal globular categories as the natural algebraic environment for defining weak n-categories; the globular tower of n-conversions in the present paper fits that framework, and the recursive completion principle of Definition 5.2 can be viewed as a free globular resolution. Batanin's work does not itself address λ-calculus, but it provides the coherence-theoretic language—globularity, strict source/target compatibility, composition by whiskering—that our construction instantiates in the λ-calculus setting.

Grothendieck's *Pursuing Stacks* [4] proposed that the fundamental ω-groupoid of a space should serve as a complete algebraic invariant. The analogy with the present work is suggestive: the space of λ-terms modulo $\beta\eta$-conversion has, by Theorem 5.6, a natural fundamental ω-groupoid whose higher cells are explicit conversion witnesses, and the 0-truncation of that groupoid recovers the classical theory (the Corollary on 0-truncation in Section 5). We do not claim a direct technical connection to Grothendieck's programme—our construction is combinatorial and syntactic rather than topological—but the conceptual alignment with the idea that "higher identification data should be explicit and algebraically coherent" is strong.

Homotopy type theory [5] provides a different realization of the same principle: identity types in Martin-Löf type theory form a weak ω-groupoid, and the univalence axiom enriches the resulting structure further. Our setting is the untyped λ-calculus rather than dependent type theory, and the higher cells are generated by reduction steps rather than by identity elimination. Nevertheless, the structural parallel is close enough that one may ask (see Open Problem 3 below) whether there is a precise translation between the λ-ω-groupoid $\mathscr{G}_\lambda$ and the fundamental ω-groupoid constructions in HoTT. More broadly, the present work connects to the logic-and-computation community's interest in proof relevance and computational interpretations of identity: where HoTT uses identity types to generate higher structure in a typed setting, we use directed reduction witnesses to generate it in the untyped λ-calculus, offering a complementary perspective on the computational content of equivalence proofs.

Lurie's *Higher Topos Theory* [7] develops a comprehensive framework for higher categories in which ∞-groupoids play a central role as the homotopy types of spaces. The relationship to the present work is at the level of conceptual architecture rather than direct technical overlap: we are constructing a concrete combinatorial ω-groupoid from the syntax of the λ-calculus, while Lurie's machinery addresses the abstract higher-categorical foundations. The two perspectives meet in the observation that the λ-ω-groupoid $\mathscr{G}_\lambda$ is an explicitly presented object of the kind that Lurie's theory classifies.

Classical domain-theoretic lineage. The K_∞ construction (Section 7) is a direct descendant of Scott's continuous lattice semantics [12]. Scott showed that every countably based continuous lattice admits a reflexive structure, and the resulting models interpret the untyped λ-calculus. The K_∞ tower replaces Scott's single-step reflexivity with an iterated function-space tower $K_{n+1} = [K_n \to K_n]$ whose inverse limit inherits both the classical algebraic-domain structure and a new homotopical layer.

Barendregt's [1] systematic treatment of λ-models provides the classical reference frame: the notions of extensionality, the $\beta\eta$-theory, Church–Rosser, and confluence that appear in our preliminaries and baseline sections are standard in the sense of [1]. The novelty of the present approach is that these classical ingredients are enriched with higher-dimensional semantic structure that was invisible in the purely propositional setting.

Nature of the contribution. The rôle of the formal development is methodological and certifying rather than expository. The mathematical burden of the paper is carried by the arguments in Sections 5–8. Among those, Theorem 7.6 is the strongest model-specific statement, Theorem 6.2 identifies the minimal coherence input needed for the later semantic comparison arguments, and Theorems 5.6 and 8.5 describe structural consequences of the recursive higher tower once the low-dimensional and point-level data are fixed. Lean 4 [6] provided a disciplined setting in which these arguments were discovered and checked, but the paper is intended to stand or fall on its mathematics rather than on the existence of the formalization. In the broader logic-and-computation tradition, the results illustrate a recurring theme: that the proof-relevant refinement of a classical equivalence relation (here, $\beta\eta$-convertibility) reveals computational structure—witness classes, coherence data, semantic separation—that the propositional collapse erases. The constructive character of the proofs, particularly the explicit witness constructions, aligns with the programme of making the computational content of logical derivations visible and tractable.

9.3. Open problems

1. **Eliminating the front-seed.** Derive the reduced front-seed coherence package (seeds FS1 and FS2 in Theorem 6.2) directly from the bare extensional Kan complex interface. A positive answer would unify the bare and coherent semantic regimes, showing that every extensional Kan complex is automatically coherent in the sense needed for associator and pentagon reasoning.
2. **Typed extensions.** Extend the all-dimensional coherence to typed λ-calculi, where the type structure could provide additional constraints on the higher-cell tower. Simply-typed and polymorphic calculi are natural first targets.
3. **Connections to HoTT.** Investigate the relationship between the λ-ω-groupoid $\mathscr{G}_\lambda$ and the fundamental ω-groupoid constructions in homotopy type theory [5]. Is there a precise functor, or at least a comparison map, between the two?

4. **Explicit strictification.** Determine whether the weak ω-groupoid $\mathscr{G}_\lambda$ admits a strictification to a strict ω-category, and if so, what information about the λ-calculus is lost in the process.

9.4. Formal verification note

All results proved in this paper have been formally verified in the Lean 4 theorem prover [6]; the complete development is available at

`https://github.com/Arthur742Ramos/HigherLambdaModel`

and contains no use of `sorry` (the Lean keyword for an unfinished proof obligation). From the mathematical point of view, the formalization plays an evidential role: it certifies that every case split, boundary compatibility check, and recursive induction invoked above has been verified mechanically. In particular, the recursive inductions in the proofs of Theorems 5.6 and 7.6 involve several dozen case-by-case verifications that are straightforward but tedious to reproduce on paper; the formal proof provides independent assurance that none were dropped. From a logic-and-computation perspective, the formalization also serves as a computational certificate: the Lean kernel's type-checking algorithm independently verifies every logical step, providing a degree of assurance that complements traditional peer review. The formalization also covers all four main theorems, the baseline results of Section 3, and the full explicit tower through dimension 6.

To keep the present paper focused on the mathematics, we record only one short code fragment in the main text—the `ProjectionPair` interface from Section 7. For readers who want a direct bridge back to the formal development, Appendix A records representative Lean entry points for the four main theorems.

A. Lean roadmap for the main theorems

This appendix gives a compact map from the four headline results of the paper to representative declarations in the Lean development, together with short illustrative excerpts. It is not an exhaustive index; the full paper-to-Lean claim matrix is maintained in `docs/theorem_index.md`. The excerpts below are intentionally short and omit some argument lists with `...` when the omitted data are not mathematically central to the discussion in the paper.

Theorem A: all-dimensional constructive coherence

Primary files.

```
HigherLambdaModel/Lambda/Coherence.lean
HigherLambdaModel/Lambda/TruncationCore.lean
HigherLambdaModel/Lambda/Truncation.lean
```

Representative declarations.

```
AllDimensionalHigherConversionCoherence
lambdaOmegaConstructiveRealize
lambdaConstructiveHigherConversionCoherence
```

Listing 2. Representative Lean excerpt for Theorem A (from `Lambda/TruncationCore.lean`).

```
def lambdaConstructiveHigherConversionCoherence :
    HigherLambdaModel.Lambda.Coherence.AllDimensionalHigherConversionCoherence
      HigherLambdaModel.Lambda.NTerms.lambdaTower
      lambdaOmegaReflexiveTower where
  realize := lambdaOmegaConstructiveRealize
  source_comm := by
    intro n x
    exact lambdaOmegaConstructiveRealize_source_comm x
  target_comm := by
    intro n x
    exact lambdaOmegaConstructiveRealize_target_comm x
```

Theorem B: reduced front-seed coherence

Primary file.

```
HigherLambdaModel/Lambda/ExtensionalKanHigher.lean
```

Representative declarations.

```
FrontSeedCoherentExtensionalKanComplex
FrontSeedCoherentExtensionalKanComplex.reductionSeq_comp_associator_in_Theory3
FrontSeedCoherentExtensionalKanComplex.reductionSeq_pentagon_in_Theory3
FrontSeedCoherentExtensionalKanComplex.homotopy2_pentagon_shell_bridge_in_Theory3
```

Listing 3. Representative Lean excerpt for Theorem B (from `Lambda/ExtensionalKanHigher.lean`).

```
structure FrontSeedCoherentExtensionalKanComplex
    extends ExtensionalKanComplex where
  pentagonInnerRightFrontReflPath3 : ...
  wlwrPath3 : ...

noncomputable def
    FrontSeedCoherentExtensionalKanComplex.reductionSeq_comp_associator_in_Theory3
    (K : FrontSeedCoherentExtensionalKanComplex) ... :=
  reductionSeq_comp_associator_in_Theory3_ofPentagonInnerRightFrontRefl
    K.toExtensionalKanComplex
    K.pentagonInnerRightFrontReflPath3
    K.wlwrPath3 ...
```

Theorem C: proof-relevant witness separation

Primary files.

```
HigherLambdaModel/KInfinity/Examples.lean
HigherLambdaModel/KInfinity/ContinuousSemantics.lean
```

Representative declarations.

```
betaEtaPaper_beta1Witness_interpretation
betaEtaPaper_eta1Witness_interpretation
betaEtaPaper_witness_interpretations_distinct
betaEtaPaper_witness_interpretations_no_recursive_higher_cell
```

Listing 4. Representative Lean excerpt for Theorem C (from `KInfinity/Examples.lean`).

```
noncomputable def betaEtaPaper_beta1Witness_interpretation :
    Example42NTerm1WitnessInterpretation :=
  example42NTerm1Witness_interpretation betaEtaPaper_beta1Witness

noncomputable def betaEtaPaper_eta1Witness_interpretation :
    Example42NTerm1WitnessInterpretation :=
  example42NTerm1Witness_interpretation betaEtaPaper_eta1Witness

theorem betaEtaPaper_witness_interpretations_no_recursive_higher_cell
    (n : Nat) :
    not (Nonempty {x : kInfinityTower.Cell (n + 5) // ... }) := ...
```

Theorem D: exact reflexive packaging of K_∞

Primary file.

```
HigherLambdaModel/KInfinity/Properties.lean
```

Representative declarations.

```
hInfinity
kInfinityContinuous
kInfinityReflexiveCHPO
applicationContinuous
```

Listing 5. Representative Lean excerpt for Theorem D (from `KInfinity/Properties.lean`).

```
noncomputable def kInfinityReflexiveCHPO : ReflexiveCHPO KInfinityCHPO where
  reify := hInfinityContinuous
  reflect := kInfinityContinuous
  retract := by
    ext x
    exact hInfinity_kInfinityContinuous_apply x

noncomputable def applicationContinuous :
    ContinuousMap (Product.chpo KInfinityCHPO KInfinityCHPO) KInfinityCHPO := ...
```

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